Maximum Practical Energy Efficiency

Detailed Implementation Handbook | Tasks 21-50

Software-first, measurement-led thesis plan for Camera-LiDAR-Radar fusion, reliable adaptive CPU/FPGA execution, dynamic precision and board-level energy control.

Planning edition | 2 November 2026 | Proposed continuation of the existing Task 21-50 roadmap. This is an optimization plan, NOT a claim of achieved minimum power.

Executive objective

Research aim: minimize measured system energy per useful perception result while meeting fixed latency and perception-quality requirements. Favor Pareto-efficient operating points, not minimum instantaneous power or maximum FPGA utilization.

Starting evidence: Task 20 corrected GEMSTAR radar range from 10 m to 50 m; recorded returns extend to about 49.93 m. The earlier sensor analysis used recorded OSI data. Radar object-association accuracy, reliable camera/LiDAR payload availability, full fusion, energy savings, and live end-to-end streaming are NOT yet independently validated.

Data + MATLAB	21-23	No	Reusable verified datasets and feature baselines
Fusion + CPU proof	24-29	No	Validated fusion, profiling, quantization feasibility, cost models
Hardware design	30-36	ARM-FPGA from 30	Efficient kernels and measured energy/latency
Adaptive optimization	37-41	ARM-FPGA	Joint placement, precision, sensor workload, clock-policy controller
Live integration	42-46	ARM-FPGA	Streaming, synchronization, monitored live switching
Thesis validation	47-50	ARM-FPGA	Statistically defensible energy + quality comparison

Non-negotiable research rules

- Do not claim 100% optimal or globally minimum energy. Report best measured results within tested hardware, workloads, deadlines, sensor-quality constraints, and power states.
- Do not equate latency with energy: measure average total power over operation plus idle/transfer costs, and integrate to Joules per frame or per correct tracked object.
- Do not drop entire safety-relevant sensors based solely on weather labels. Use explicit minimum sensing/quality thresholds, stale-data detection and fallback.
- Use fair identical input recordings/streams and identical output requirements across CPU-only, fixed FPGA, adaptive placement, and full joint controller.
- Maintain immutable raw OSI recordings and traceable timestamp, map, vehicle, weather, configuration, firmware, compiler and software revisions.

Energy model and optimization objective

Measure E_system = integral P_board(t) dt over a matched evaluation window; E_frame = (E_run - E_idle_reference, when reported as incremental energy) / completed valid frames. Always report BOTH total system energy and incremental workload energy, with the idle reference and integration boundary explicitly documented.

Joint decision u(t) = {sensor rate/resolution/ROI, preprocessing fidelity, numeric precision, CPU/FPGA placement, accelerator enable, supported clock/power state}. Objective: minimize measured or calibrated predicted Joules per valid output subject to p99 latency <= deadline, quality >= threshold, acceptable missed detections, bounded queues, and thermal constraints.

The accuracy and safety thresholds must be defined before experiments with a supervisor; do not infer automotive safety certification from simulation results.

Reference architecture

GEMSTAR scenes -> timestamped OSI data or live sensor transport -> sensor health / scene complexity -> deadline and quality guards -> constrained policy -> ARM tasks and resident FPGA streaming kernels -> fusion/tracking -> monitoring and logged outcomes. ARM handles supervisory control, fusion and fallback; FPGA targets repeated streaming kernels only where total-system energy and throughput justify acceleration.

Recommended optimization hierarchy

- First remove unnecessary computation through validated region-of-interest, point thinning, voxel size, adaptive resolution and frame budgets.
- Then optimize memory/data movement: zero-copy where possible, burst DMA, contiguous buffers, on-chip reuse, stream processing, avoiding CPU-FPGA round-trips.
- Then apply measured bit-width reduction (fixed-point / INT8 where acceptable), pipeline II=1 when feasible, DSP/BRAM sharing and clock enables.
- Then deploy calibrated dynamic placement and supported operating modes; keep hysteresis and transition cost in decisions.
- Only evaluate expensive additions such as dynamic partial reconfiguration or deep RL after a simple constrained baseline demonstrates need.

Detailed task-by-task implementation roadmap

Every task must end in tests/Task_XX_Final_Research with code, raw-data references, CSV outputs, figures, configuration, findings and a concise research report. Keep original recordings immutable.

Phase A | MATLAB and sensor data

Task 21 - MATLAB radar analysis and clean import

- **1.** Create Task_21_Final_Research folders and import Task 20 radar CSVs; retain source hashes.
- **2.** Reproduce range, signal, frame and startup-artifact plots in MATLAB; label unverified ray-to-object association.
- **3.** Time CSV load, vectorized filters and feature extraction with warm-up and repeated trials.

Deliverable: Radar analytics CSVs, MATLAB scripts, figures and timing distributions.

Exit gate: All numbers reproducible from source CSVs; no radar accuracy claim.

Task 22 - Map-agnostic and scenario-controlled OSI pipeline

- **1.** Parameterize paths, frame ranges, OSI schema/version and scenario metadata; remove fixed vehicle IDs.
- **2.** Add missing-payload/timestamp, units, coordinate and host pose checks; verify cross-map geometry.
- **3.** Collect paired clear/fog/dense-traffic and second-map recordings with documented simulation seeds.

Deliverable: Reusable loader, configuration files, dataset manifests and integrity audit.

Exit gate: Each recording produces schema/status report; changes of map need no script edits.

Task 23 - Camera/LiDAR/Radar preprocessing readiness

- **1.** Verify actual camera and LiDAR payloads before assuming tri-sensor availability.
- **2.** Build comparable CPU features: camera edges/ROI, LiDAR voxel/downsample, radar return filtering.
- **3.** Test degradation, corrupt samples, clocks and invalid inputs with ground-truth-informed checks.

Deliverable: Three sensor preprocessing modules and quality/coverage reports.

Exit gate: Payloads available and usable; ground-truth and coordinates independently checked.

Phase B | Fusion and CPU reference

Task 24 - Three-sensor association and tracking fusion

- **1.** Calibrate transforms, coordinate conventions and time alignment.
- **2.** Start object-level association plus EKF tracking, with confidence and outlier gating.
- **3.** Compare camera-only, LiDAR-only, radar-only and fused results on labelled paired scenarios.

Deliverable: Tracked-object CSVs, metrics, MATLAB reference and Python implementation.

Exit gate: Validation demonstrates meaningful object association and stable uncertainty.

Task 25 - Fusion reliability and failure constraints

- **1.** Implement per-sensor health from missing, fog-like degradation, stale messages and innovation residuals.
- **2.** Quantify recall/precision or tracking error, association errors, false positives and latency.
- **3.** Define supervisor-agreed failure rules, minimum quality and fallback.

Deliverable: Reliability score calibration and locked acceptance metrics.

Exit gate: Thresholds fixed before hardware comparisons; fallback tested.

Task 26 - Fair software CPU benchmark

- **1.** Pin input data, algorithm versions, resolution and output conditions.
- **2.** Measure warm-up, p50/p95/p99 latency, throughput, CPU use, RAM and transfer-independent kernel time.
- **3.** Record measurement environment, process affinity, thermals and potential OS jitter.

Deliverable: CPU reference outputs, timings, audit-ready baseline.

Exit gate: Repeated runs stable; output equivalence criteria defined.

Phase C | Design-space selection

Task 27 - Workload-level energy and roofline diagnosis

- **1.** Profile image, point cloud, radar, fusion and IO as separate components.
- **2.** Measure bytes moved, operations, working set, queuing and memory-bound sections.
- **3.** Estimate best hardware candidates from attainable bandwidth/compute, not intuition.

Deliverable: Prioritized workload ranking with transfer size and expected acceleration break-even.

Exit gate: Select at most two initial hardware kernels.

Task 28 - Quantization/fixed-point feasibility

- **1.** Analyze value ranges for voxel, image and radar features; choose Q formats or INT8 selectively.
- **2.** Run floating vs reduced precision with saturation/rounding and worst-case scenarios.
- **3.** Measure actual algorithm-level quality and resource proxy; reject precision modes that violate bounds.

Deliverable: Numerical precision report and test vectors.

Exit gate: Quality thresholds maintained with margin across scenarios.

Task 29 - Joint scheduling specification and simulator

- **1.** Build an offline discrete-event simulator for CPU/FPGA task, DMA and queue costs.
- **2.** Define constrained actions including rate/ROI/precision/placement/power state; add switching penalties.
- **3.** Benchmark always-CPU, fixed-offload, simple heuristic and idealized lower-bound oracle.

Deliverable: Controller state/action schema, constrained policy baseline and trace logger.

Exit gate: Do not claim optimality from an uncalibrated energy model.

Phase D | Hardware acquisition and bring-up

Task 30 - ARM-FPGA platform bring-up and measurement contract

- **1.** Select SoC board with ARM cores, available DDR/DMA/AXI, suitable dev tools and measurable rails.
- **2.** Implement reproducible toolchain, bitstream deployment, hello-world DMA/AXI test and timestamps.
- **3.** Choose credible external/instrumented power measurement path; document sensor, sample rate and accuracy.

Deliverable: Working platform, board specification, instrumentation diagram, idle-power baseline.

Exit gate: Real board transfer succeeds; measured power instrumentation is operational.

Phase E | Hardware acceleration and measurement

Task 31 - DMA-first zero-copy-friendly streaming path

- **1.** Design contiguous host buffers, AXI DMA descriptors, burst sizes and cache coherence.
- **2.** Measure input copy, transfer, kernel, interrupt, queue and output copy separately.

- **3.** Sweep batches and payload sizes; find actual CPU-vs-offload break-even.

Deliverable: Transfer microbenchmarks, frame deadlines and DMA log.

Exit gate: No hidden transfer costs in hardware comparisons.

Task 32 - LiDAR FPGA voxel/downsample accelerator

- **1.** Implement synthesizable streaming point filter and voxel/downsample candidate; preserve needed geometry.
- **2.** Use fixed point where validated, bounded BRAM and deterministic overflow policy.
- **3.** Compare quality, total wall time and actual energy to optimized ARM equivalent.

Deliverable: RTL/HLS source, testbench, implementation reports, actual board measurements.

Exit gate: Equivalent output/quality and real Joules-per-frame benefit on chosen operating points.

Task 33 - Camera FPGA preprocessing accelerator

- **1.** Implement streaming grayscale/edges/ROI or selected compact INT8 features.
- **2.** Avoid full-frame DDR round trips if streaming allows; verify borders, precision and throughput.
- **3.** Measure FPGA activity, idle, power and end-to-end effect on fusion.

Deliverable: Image accelerator and fidelity/energy plots.

Exit gate: Image accuracy, end-to-end deadlines and bus bandwidth meet targets.

Task 34 - Radar hardware kernel feasibility gate

- **1.** Evaluate radar return filtering/range conversion only if CPU profile and DMA crossover justify it.
- **2.** Otherwise retain radar on ARM and invest FPGA resources in two proven accelerators.
- **3.** If built, verify units, threshold behavior, precision and frame alignment.

Deliverable: Documented keep-on-CPU or offload decision with data.

Exit gate: Do not add a weak accelerator merely for symmetry.

Task 35 - Board-level power and energy measurements

- **1.** Measure idle, CPU-only, FPGA-resident-idle, kernel, DMA, concurrent and full-frame intervals.
- **2.** Capture rail/system power at sufficient sampling resolution; align time and workload markers.
- **3.** Report total board J/frame, dynamic-minus-idle J/frame, p95/p99 latency and thermal conditions.

Deliverable: Power trace CSV, calibration, repeat tests, uncertainties and energy curves.

Exit gate: Measured energy, not a guessed wattage or tool-estimated power alone.

Task 36 - Streaming pipeline and power optimization

- **1.** Sweep II, resource reuse, BRAM buffering, DMA burst, batch size and clock enables.
- **2.** Minimize spurious switching; test supported frequency and power controls safely.
- **3.** Choose Pareto frontier from measured energy, latency and fidelity, including static power.

Deliverable: Pareto plots and finalized accelerator operating modes.

Exit gate: No deadline/quality regression; measurable improvement from at least one technique.

Phase F | Energy-aware control

Task 37 - Calibrated sensor health and workload prediction

- **1.** Extract camera confidence, LiDAR returns, radar consistency, object density and queue signals.
- **2.** Calibrate predictors across maps; compare sensor-health labels with truth/degradation.

- **3.** Quantify uncertainty, false alarms and computational overhead.

Deliverable: Health/scene predictor with calibration and cross-map evaluation.

Exit gate: Policy cannot blindly trust inaccurate confidence scores.

Task 38 - Hardware-aware energy/latency prediction

- **1.** Fit small per-mode models using measured board runs including CPU/FPGA switching/transfer costs.
- **2.** Validate models on withheld scenarios and workload levels.
- **3.** Bound prediction error conservatively; use observed worst-case costs near deadlines.

Deliverable: Calibrated execution-cost lookup or lightweight estimator.

Exit gate: Holdout prediction error and uncertainty documented.

Task 39 - Joint constrained adaptive scheduler

- **1.** Solve/select placement + precision + ROI/rate + operating mode within deadline/quality guards.
- **2.** Implement lightweight constrained rule/lookup optimizer first; add hysteresis and cooldown.
- **3.** Log decisions, candidate cost, fallback reason, overhead and actual outcome.

Deliverable: Realtime policy code, logs, schedule trace and reproducible scenarios.

Exit gate: Policy overhead < benefit; deadline constraints enforced.

Task 40 - Safe power-state and precision switching

- **1.** Add supported accelerator enables/clock control and precision-mode changes only on valid boundaries.
- **2.** Measure wake-up/transition energy, dropped/stale frames and clock-domain correctness.
- **3.** Reject unsafe state transitions; fallback to known-good mode.

Deliverable: Transition measurements and state-machine tests.

Exit gate: No corrupted outputs or unacceptable missed deadlines during transitions.

Task 41 - Scheduler ablations and robustness

- **1.** Compare fixed placement, placement-only, +precision, +workload budget, +power control, full controller.
- **2.** Test noisy health estimates, startup, queue spikes, sensor dropout, fog and mixed scenes.
- **3.** Report ablation deltas in J/frame, p99 latency and quality with intervals.

Deliverable: Evidence that each feature matters or does not.

Exit gate: Full controller beats strongest fair baseline on claimed metrics.

Phase G | Live processing and scenarios

Task 42 - Live GEMSTAR streaming transport

- **1.** Implement publisher/receiver from existing OSI replay first, then actual live GEMSTAR sensor path.
- **2.** Track sequence, timestamps, packet drops, queuing, jitter and backpressure.
- **3.** Keep archive/replay switch for deterministic debugging.

Deliverable: Live transport, logs and replay reproducer.

Exit gate: Measured actual live stream, not rebranded replay.

Task 43 - OSI-to-ROS2 topics and timestamp synchronization

- **1.** Publish camera/image, point-cloud and radar messages with frame IDs and transforms.
- **2.** Compare event time, reception time and simulated wall time; implement sync windows.

- **3.** Validate message loss handling and latency budgets.

Deliverable: ROS2 topic graph, recorded samples and sync plots.

Exit gate: No silently mixed timestamps or transforms.

Task 44 - Live hardware processing and fusion

- **1.** Connect streams through ARM preprocessor, FPGA DMA and resident kernels to fusion.
- **2.** Stress concurrent camera/LiDAR workloads and return-path bandwidth.
- **3.** Maintain full end-to-end trace IDs for each object/frame.

Deliverable: Live fused tracker with transparent CPU/FPGA trace.

Exit gate: Runs within defined p99 latency and acceptable quality.

Task 45 - Continuous clear-fog-dense traffic scenario

- **1.** Prepare one uninterrupted GEMSTAR scenario with controlled transition markers.
- **2.** Validate weather model effects and synthetic degradation plausibility.
- **3.** Monitor adaptation, thermal, buffering and perception throughout transitions.

Deliverable: Continuous demo log, scenario metadata, raw streams and video.

Exit gate: All three scenarios captured without undisclosed resets.

Task 46 - Live research dashboard

- **1.** Display sensor health, processing location, precision, clock mode and fallback events.
- **2.** Display measured total J/frame/average W, latency distributions, queue and quality metrics.
- **3.** Audit visualization vs original time-series data.

Deliverable: Live dashboard plus exportable CSV and demo recording.

Exit gate: Dashboard values trace directly to measured logs.

Phase H | Thesis-grade evidence

Task 47 - Pre-registered experimental matrix

- **1.** Fix scenes, seeds, repetitions, modes and quality/latency thresholds prior to measurement.
- **2.** Run CPU, fixed FPGA, adaptive and ablated policies on matched workloads.
- **3.** Repeat across clear/fog/traffic and map(s), including failure cases.

Deliverable: Protocol, raw experiment matrix, versioned configurations.

Exit gate: Comparisons are fair and reproducible.

Task 48 - Statistical analysis and Pareto findings

- **1.** Calculate total J/frame, incremental J/frame, p50/p95/p99 latency, throughput and quality.
- **2.** Report variation/confidence intervals across runs and seed sensitivity.
- **3.** Separate transfer, idle, accelerator and CPU energy; quantify tradeoffs honestly.

Deliverable: Tables, Pareto frontier, uncertainty and failure analysis.

Exit gate: Any energy saving supported by measured board data.

Task 49 - Thesis evidence package and limitations

- **1.** Consolidate RTL/HLS, Python, MATLAB, configs, OSI manifests, power traces and CSVs.
- **2.** Document novelty against prior research and reproducibility boundaries.

- **3.** State unvalidated safety, weather realism and hardware portability limits.

Deliverable: Research archive, methods/results manuscript and advisor-review checklist.

Exit gate: All claims auditable from a versioned artifact.

Task 50 - Final live demonstration and thesis submission

- **1.** Show continuous clear -> fog -> traffic input with sensor health and scheduler decisions.
- **2.** Show CPU-only, fixed and adaptive comparisons including energy and p99 latency.
- **3.** Finish supervisor revisions, figures, reproducibility annex and defense demo.

Deliverable: Final thesis, reproducible research package and live video.

Exit gate: Quality/deadline guard met; claims limited to tested setup.

Benchmarking protocol: measure actual minimum attainable energy

- Baseline A: optimized ARM-only software; baseline B: resident fixed FPGA offload; baseline C: adaptive placement; baseline D: full joint energy controller.
- Use same raw inputs, frame budgets, detection/tracking output semantics and deadlines. Record any dropped frames, processing-rate changes or quality deterioration.
- Measure total-system average W and Joules per valid frame; additionally report idle-subtracted energy with its reference. Battery/external sensor power should be included only if comparable and measurable.
- Separate end-to-end timestamp latency from kernel runtime. Report p50/p95/p99, misses, throughput, DMA/queue overhead and policy overhead.
- To maximize efficiency in tested conditions, sweep allowed clock, data width, ROI/voxel resolution, batch size, DMA modes and idle states and choose non-dominated operating points.
- Use at least several repeat runs and multiple seeds; quantify uncertainty and do not extrapolate to all driving conditions.

Example acceptance specification (to be tuned before experiments)

Perception fidelity	Predefine tracking/detection metric and maximum permitted degradation vs accurate reference
Latency deadline	Application-defined bound; report p99 and misses, not just mean
Energy efficiency	Report total J/frame and % reduction against both CPU and optimized fixed FPGA
Quality-preserving savings	Only count energy improvements when quality/deadline constraints are satisfied
Scheduler behavior	No rapid oscillation; show decisions and transition penalties
Evidence strength	Repeatable experiment matrix, raw power traces, source version and uncertainty

Risk register and mitigations

Camera/LiDAR payload unavailable	Block fusion claims until sensor output validated; use recorded ground-truth-independent diagnostics

Radar ray misses external targets	Controlled target scenes, verified actor hit IDs, full geometry/time alignment
FPGA adds idle power	Report board static power; suspend/disable processing only if supported and beneficial
DMA dominates small kernels	Determine transfer break-even; retain CPU path for short workloads
Quantization hurts fidelity	Precision error tests with saturation checks; selective fallback
Predictor wrong in fog	Guard conditions and robust conservative fallback; monitor uncertainty
Live stream overload	Bounded queues, admission control, timestamp monitoring and reproducible replay
Hardware/software validation delayed	Ship validated two-accelerator minimum viable thesis instead of three half-validated kernels

Recommended minimum viable implementation

Build and prove two streaming accelerators (LiDAR filtering and camera edges/ROI), CPU EKF fusion/tracking, one measured power interface, CPU-only and fixed-FPGA baselines, and a constrained scheduler that changes placement/precision/workload budgets based on calibrated sensor health. Radar stays on CPU unless profiling demonstrates real value for hardware. This avoids unnecessary implementation risk while preserving a strong and testable contribution.

Final scientific claims you MAY make after successful measurement

- Under specified board, scenes, loads and thresholds, full controller reduced total measured board Joules/frame by X% compared with fixed FPGA and Y% compared with CPU-only.
- Under the same acceptable perception quality, p99 latency was below D ms, with N deadline misses across M evaluated frames.
- The ablation study attributes Z percentage points of savings to workload budgeting or clock enables, with measurement uncertainty.

Do NOT claim globally minimum power, certified automotive safety, universally optimal AI scheduling, or validated energy savings until the corresponding evidence exists.

Project documentation locations

Suggested root: E:\GEMSTAR_Project\sensor-health-monitor\tests\Task_XX_Final_Research\. Required subdirectories: 01_Code, 02_CSV_Data, 03_Configurations, 04_Plots, 05_Raw_Data_References, 06_Research_Findings, 07_Research_Report, 08_Power_Traces (hardware tasks). Use the existing Task 20 archive as a reference, not as a dataset to overwrite.

Authoring note

Task sequence is a forward implementation proposal based on the completed Task 20 radar investigation and the existing Task 21-50 roadmap. Choices of board, sensor data rates, allowable precision, power modes, actual clock gating and energy targets depend on later hardware and benchmark results. No savings percentage is guaranteed.